

Hardware information

Please do not backdrive the motors! The motion controllers do not have brake choppers included.

The lab hardware consists of motors, gear and motion controllers from Faulhaber. The following hardware is used:

Single motor configuration

- Drives of end effector:
 - Motor: 2232S024BX4
 - 4-stage planetary gears: 22GPT with total ratio 323:1
 - Speed and position control by hall sensor values.
- Motion Controller: MC 3603 S ET

Robot arm

- Drives of axis 1 to 3
 - Motor: 3216X024BXTH
 - 3-stage planetary gears: 32GPT with total ratio 196:1
 - Encoder: IEF3-4096
 - Speed and position control by encoder values.
- Drives of end effector:
 - Motor: 2232S024BX4
 - 4-stage planetary gears: 22GPT with total ratio 323:1
 - Speed and position control by hall sensor values.
- Motion controllers: MC 5004 P ET

To determine the right number of increments per rotation, please check out how incremental encoders by Faulhaber work and how the motion controllers use analog hall sensors signals as position encoders.